

*RETRIEVAL

*DMLIST

*NO-ACTIVITY-LOG

*SCHEMA-COMMENTS

IDENTIFICATION DIVISION.

PROGRAM-ID. DEPTRPT.

AUTHOR. CA.

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REMARKS. THIS PROGRAM DEMONSTRATES
CA IDMS DATABASE ACCESS USING
COBOL DML STATEMENTS. IT READS
DEPARTMENT ID NUMBERS AND RETRIEVES
RELATED RECORD OCCURRENCES,
PRINTING A REPORT THAT INCLUDES
DEPARTMENT, EMPLOYEE, JOB, AND
OFFICE INFORMATION.

ENVIRONMENT DIVISION.

INPUT-OUTPUT SECTION.

FILE-CONTROL.

SELECT DEPT-FILE-IN ASSIGN TO INFILE.

SELECT DEPT-FILE-OUT ASSIGN TO OUTFILE.

SELECT ERR-FILE-OUT ASSIGN TO ERRFILE.

IDMS-CONTROL SECTION.

PROTOCOL. MODE IS BATCH DEBUG

IDMS-RECORDS MANUAL.

SKIP3

DATA DIVISION.

SCHEMA SECTION.

DB EMPSS01 WITHIN EMPSCHM.

FILE SECTION.

FD DEPT-FILE-IN
RECORD CONTAINS 80
BLOCK CONTAINS 80 CHARACTERS
RECORDING MODE IS F
LABEL RECORDS ARE OMITTED.

01 DEPT-REC-IN.
02 DEPT-ID-IN PIC 9(4).
02 DEPT-IN-FILLER PIC X(76).

FD DEPT-FILE-OUT
RECORD CONTAINS 133
BLOCK CONTAINS 133 CHARACTERS
RECORDING MODE IS F
LABEL RECORDS ARE OMITTED.

01 DEPT-REC-OUT.
02 CC PIC X.
02 PRINT-LINE PIC X(132).

FD ERR-FILE-OUT
RECORD CONTAINS 133
BLOCK CONTAINS 133 CHARACTERS
RECORDING MODE IS F
LABEL RECORDS ARE OMITTED.

01 ERR-REC-OUT.
02 ERR-CC PIC X.
02 ERR-LINE PIC X(132).

WORKING-STORAGE SECTION.

01 EOF-SW PIC X VALUE 'N'.
88 END-OF-FILE VALUE 'Y'.
01 LINE-COUNT PIC 99 VALUE 0.
01 ERR-LINE-COUNT PIC 99 VALUE 0.
01 LINE-MAX PIC 99 VALUE 50.

01 DEPT-HEADER.
05 FILLER PIC X(30) VALUE SPACES.
05 FILLER PIC X(13) VALUE 'DEPARTMENT ID'.
05 FILLER PIC X(10) VALUE SPACES.
05 FILLER PIC X(9) VALUE 'DEPT NAME'.
05 FILLER PIC X(70) VALUE SPACES.

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01 DEPT-DETAIL-LINE.
  05 FILLER  PIC X(33) VALUE SPACES.
  05 DEPT-ID-OUT  PIC X(4).
  05 FILLER  PIC X(16) VALUE SPACES.
  05 DEPT-NAME-OUT  PIC X(45).
  05 FILLER  PIC X(34) VALUE SPACES.
01 EMP-HEADER.
  05 FILLER  PIC X(5)  VALUE SPACES.
  05 FILLER  PIC X(6)  VALUE 'EMP ID'.
  05 FILLER  PIC X(2)  VALUE SPACES.
  05 FILLER  PIC X(9)  VALUE 'LAST NAME'.
  05 FILLER  PIC X(8)  VALUE SPACES.
  05 FILLER  PIC X(10) VALUE 'FIRST NAME'.
  05 FILLER  PIC X(3)  VALUE SPACES.
  05 FILLER  PIC X(10) VALUE 'START DATE'.
  05 FILLER  PIC X(2)  VALUE SPACES.
  05 FILLER  PIC X(9)  VALUE 'JOB TITLE'.
  05 FILLER  PIC X(13) VALUE SPACES.
  05 FILLER  PIC X(14) VALUE 'OFFICE ADDRESS'.
  05 FILLER  PIC X(42) VALUE SPACES.
01 EMP-DETAIL-LINE.
  05 FILLER  PIC X(5)  VALUE SPACES.
  05 ID-OUT  PIC X(4).
  05 FILLER  PIC X(4)  VALUE SPACES.
  05 LAST-OUT  PIC X(15).
  05 FILLER  PIC X(2)  VALUE SPACES.
  05 FIRST-OUT  PIC X(10).
  05 FILLER  PIC X(3)  VALUE SPACES.
  05 SD-OUT.
    10 SD-MM  PIC XX.
    10 FILLER  PIC X   VALUE '/'.
    10 SD-DD  PIC XX.
    10 FILLER  PIC X   VALUE '/'.
    10 SD-YY  PIC XX.
  05 FILLER  PIC X(4)  VALUE SPACES.
  05 TITLE-OUT  PIC X(20).
  05 FILLER  PIC X(2)  VALUE SPACES.
  05 OFF-ADDRESS-OUT.
    10 STREET-OUT  PIC X(20).
    10 FILLER  PIC XX  VALUE SPACES.
    10 CITY-OUT  PIC X(15).
    10 FILLER  PIC XX  VALUE SPACES.
    10 STATE-OUT  PIC XX.
    10 FILLER  PIC XX  VALUE SPACES.

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    10 ZIP-OUT PIC X(5).
    05 FILLER PIC X(8) VALUE SPACES.
01 ERR-HEADER-1.
    05 FILLER PIC X(40) VALUE SPACES.
    05 FILLER PIC X(12) VALUE 'ERROR REPORT'.
    05 FILLER PIC X(80) VALUE SPACES.
01 ERR-HEADER-2.
    05 FILLER PIC X(10) VALUE SPACES.
    05 FILLER PIC X(4) VALUE '*** '.
    05 FILLER PIC X(51) VALUE
        'THIS REPORT LISTS EMPTY AND NONEXISTENT DEPARTMENTS'.
    05 FILLER PIC X(4) VALUE ' ***'.
    05 FILLER PIC X(63) VALUE SPACES.
01 ERR-HEADER-3.
    05 FILLER PIC X(20) VALUE SPACES.
    05 FILLER PIC X(7) VALUE 'DEPT ID'.
    05 FILLER PIC X(9) VALUE SPACES.
    05 FILLER PIC X(7) VALUE 'MESSAGE'.
    05 FILLER PIC X(89) VALUE SPACES.
01 ERR-DETAIL-LINE.
    05 FILLER PIC X(20) VALUE SPACES.
    05 ERR-ID-OUT PIC X(4).
    05 FILLER PIC X(12) VALUE SPACES.
    05 ERR-MESS-OUT PIC X(15).
    05 FILLER PIC X(79) VALUE SPACES.

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01 MESSAGES.
    05 NO-JOB-MESSAGE.
        10 FILLER PIC X(20) VALUE 'NO JOB ASSIGNED'.
    05 NO-OFFICE-MESSAGE.
        10 FILLER PIC X(20)
            VALUE 'NO OFFICE ASSIGNED'.
    05 NO-DEPT-MESSAGE.
        10 FILLER PIC X(15) VALUE 'DOES NOT EXIST'.
    05 NO-EMP-MESSAGE.
        10 FILLER PIC X(15) VALUE 'IS EMPTY'.
    05 NO-INPUT-MESSAGE.
        10 FILLER PIC XX VALUE SPACES.
        10 FILLER PIC X(11) VALUE '=====>> '.
        10 FILLER PIC X(8) VALUE 'NO INPUT'.
        10 FILLER PIC X(11) VALUE '<<====='.
        10 FILLER PIC X(100) VALUE SPACES.

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01 COPY IDMS SUBSCHEMA-CTRL.

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01 COPY IDMS SUBSCHEMA-SSNAME.

01 COPY IDMS SUBSCHEMA-RECNames.

01 COPY IDMS SUBSCHEMA-SETNames.

01 COPY IDMS RECORD EMPLOYEE.

01 COPY IDMS RECORD DEPARTMENT.

01 COPY IDMS RECORD JOB.

01 COPY IDMS RECORD EMPOSITION.

01 COPY IDMS RECORD OFFICE.

EJECT
PROCEDURE DIVISION.

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* *****
* * PROCEDURE DIVISION GENERAL STRATEGY: *
* * 1) READ DEPT-ID-IN, WHICH CONTAINS THE *
* *   DEPT-ID NUMBER *
* * 2) ACCESS THE DATABASE USING THE DEPT-ID NUMBER *
* *   WITH AN OBTAIN CALC ON THE DEPARTMENT RECORD *
* * 3) ACCESS ALL EMPLOYEES IN THE DEPT-EMPLOYEE SET *
* *   AND RETRIEVE RELATED JOB AND OFFICE DATA *
* * 4) PRINT A REPORT FOR EACH DEPARTMENT *
* * 5) PRINT AN ERROR REPORT FOR EMPTY DEPARTMENTS *
* *   AND NONEXISTENT DEPARTMENTS (NO MATCHING *
* *   DEPT-ID) *
* *****
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MAIN-LINE.

PERFORM INIT-FILES.

IF END-OF-FILE

PERFORM EMPTY-INPUT-PROCESSING

ELSE

PERFORM INIT-BIND-READY

PERFORM U220-ERR-HEADER

PERFORM DEPT-PROCESSING THRU DEPT-PROCESSING-EXIT
UNTIL END-OF-FILE.

PERFORM END-PROCESSING.

GOBACK.

INIT-BIND-READY.

* THE BIND STATEMENTS ARE PERFORMED INDIVIDUALLY (RATHER *
* THAN BY USING A COPY IDMS SUBSCHEMA-BINDS) IN ORDER TO *
* CHECK EACH ERROR-STATUS BY PERFORMING THE IDMS-STATUS *
* ROUTINE. *

MOVE 'DEPTRPT' TO PROGRAM-NAME.

BIND RUN-UNIT.

PERFORM IDMS-STATUS.

BIND EMPLOYEE.

PERFORM IDMS-STATUS.

BIND DEPARTMENT.

PERFORM IDMS-STATUS.

BIND JOB.

PERFORM IDMS-STATUS.

BIND EMPOSITION.

PERFORM IDMS-STATUS.

BIND OFFICE.

PERFORM IDMS-STATUS.

READY.

PERFORM IDMS-STATUS.

INIT-FILES.

OPEN INPUT DEPT-FILE-IN.

OPEN OUTPUT DEPT-FILE-OUT.

OPEN OUTPUT ERR-FILE-OUT.

MOVE SPACES TO PRINT-LINE.

MOVE SPACES TO ERR-LINE.

READ DEPT-FILE-IN AT END MOVE 'Y' TO EOF-SW.

EMPTY-INPUT-PROCESSING.

MOVE NO-INPUT-MESSAGE TO PRINT-LINE.

MOVE '1' TO CC.

PERFORM U000-WRITE-LINE.

* THIS PARAGRAPH ACCESSES THE DATABASE USING THE DEPT-ID-0415 *
* CALCKEY VALUE. *

DEPT-PROCESSING.

MOVE DEPT-ID-IN TO DEPT-ID-0410.

OBTAIN CALC DEPARTMENT.

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IF DB-REC-NOT-FOUND THEN
  PERFORM NO-DEPT-PROCESSING
ELSE
  PERFORM IDMS-STATUS
  IF DEPT-EMPLOYEE IS NOT EMPTY THEN
    PERFORM U020-VALID-HEADER
    MOVE DEPT-ID-0410 TO DEPT-ID-OUT
    MOVE DEPT-NAME-0410 TO DEPT-NAME-OUT
    MOVE DEPT-DETAIL-LINE TO PRINT-LINE
    PERFORM U000-WRITE-LINE
    PERFORM U030-EMP-HEADERS
    PERFORM SET-WALK THRU SET-WALK-EXIT
      UNTIL DB-END-OF-SET
  ELSE
    PERFORM EMPTY-SET.
  READ DEPT-FILE-IN AT END MOVE 'Y' TO EOF-SW.
DEPT-PROCESSING-EXIT.
EXIT.
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* THIS PARAGRAPH RETRIEVES EMPLOYEE, JOB, AND OFFICE DATA *

* FOR EACH EMPLOYEE IN THE DEPT-EMPLOYEE SET. *

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SET-WALK.
OBTAIN NEXT EMPLOYEE WITHIN DEPT-EMPLOYEE.
IF DB-END-OF-SET
  GO TO SET-WALK-EXIT
ELSE
  PERFORM IDMS-STATUS.
  MOVE EMP-ID-0415 TO ID-OUT.
  MOVE EMP-LAST-NAME-0415 TO LAST-OUT.
  MOVE EMP-FIRST-NAME-0415 TO FIRST-OUT.
  MOVE START-YEAR-0415 TO SD-YY.
  MOVE START-MONTH-0415 TO SD-MM.
  MOVE START-DAY-0415 TO SD-DD.
  IF EMP-EMPOSITION IS EMPTY
    MOVE NO-JOB-MESSAGE TO TITLE-OUT
  ELSE
    FIND FIRST WITHIN EMP-EMPOSITION
    PERFORM IDMS-STATUS
    IF NOT JOB-EMPOSITION MEMBER
      MOVE NO-JOB-MESSAGE TO TITLE-OUT
    ELSE
      OBTAIN OWNER WITHIN JOB-EMPOSITION
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PERFORM IDMS-STATUS
MOVE TITLE-0440 TO TITLE-OUT.
IF OFFICE-EMPLOYEE IS EMPTY
MOVE NO-OFFICE-MESSAGE TO STREET-OUT
MOVE SPACES TO CITY-OUT
MOVE SPACES TO STATE-OUT
MOVE SPACES TO ZIP-OUT
ELSE
OBTAIN OWNER WITHIN OFFICE-EMPLOYEE
PERFORM IDMS-STATUS
MOVE OFFICE-STREET-0450 TO STREET-OUT
MOVE OFFICE-CITY-0450 TO CITY-OUT
MOVE OFFICE-STATE-0450 TO STATE-OUT
MOVE OFFICE-ZIP-FIRST-FIVE-0450 TO ZIP-OUT
MOVE EMP-DETAIL-LINE TO PRINT-LINE.
PERFORM U000-WRITE-LINE.
SET-WALK-EXIT.
EXIT.

END-PROCESSING.
FINISH.
PERFORM IDMS-STATUS.
CLOSE DEPT-FILE-OUT.
CLOSE ERR-FILE-OUT.
CLOSE DEPT-FILE-IN.

EMPTY-SET.
MOVE SPACES TO ERR-LINE.
MOVE DEPT-ID-0410 TO ERR-ID-OUT.
MOVE NO-EMP-MESSAGE TO ERR-MESS-OUT.
MOVE ERR-DETAIL-LINE TO ERR-LINE.
PERFORM U200-WRITE-ERR-LINE.

NO-DEPT-PROCESSING.
MOVE DEPT-ID-IN TO ERR-ID-OUT.
MOVE NO-DEPT-MESSAGE TO ERR-MESS-OUT.
MOVE ERR-DETAIL-LINE TO ERR-LINE.
PERFORM U200-WRITE-ERR-LINE.

U000-WRITE-LINE.
WRITE DEPT-REC-OUT AFTER POSITIONING CC.
IF CC = '1' THEN MOVE 0 TO LINE-COUNT
ELSE IF CC = ' ' THEN ADD 1 TO LINE-COUNT
ELSE IF CC = '0' THEN ADD 2 TO LINE-COUNT.


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IF LINE-COUNT > LINE-MAX
  THEN PERFORM U010-NEW-PAGE-ROUTINE.
U010-NEW-PAGE-ROUTINE.
  PERFORM U020-VALID-HEADER.
  MOVE DEPT-DETAIL-LINE TO PRINT-LINE.
  PERFORM U000-WRITE-LINE.
  PERFORM U030-EMP-HEADERS.
U020-VALID-HEADER.
  MOVE DEPT-HEADER TO PRINT-LINE.
  MOVE '1' TO CC.
  PERFORM U000-WRITE-LINE
  MOVE '' TO CC.
U030-EMP-HEADERS.
  MOVE '0' TO CC.
  MOVE EMP-HEADER TO PRINT-LINE.
  PERFORM U000-WRITE-LINE.
  MOVE SPACES TO PRINT-LINE.
  MOVE '' TO CC.
  PERFORM U000-WRITE-LINE.

U200-WRITE-ERR-LINE.
  WRITE ERR-REC-OUT AFTER POSITIONING ERR-CC.
  IF ERR-CC = '1' THEN MOVE 0 TO ERR-LINE-COUNT
  ELSE IF ERR-CC = '' THEN ADD 1 TO ERR-LINE-COUNT
  ELSE IF ERR-CC = '0' THEN ADD 2 TO ERR-LINE-COUNT.
  IF ERR-LINE-COUNT > LINE-MAX THEN
    PERFORM U220-ERR-HEADER.
U220-ERR-HEADER.
  MOVE ERR-HEADER-1 TO ERR-LINE.
  MOVE '1' TO ERR-CC.
  PERFORM U200-WRITE-ERR-LINE
  MOVE '0' TO ERR-CC.
  MOVE ERR-HEADER-2 TO ERR-LINE.
  PERFORM U200-WRITE-ERR-LINE.
  MOVE ERR-HEADER-3 TO ERR-LINE.
  PERFORM U200-WRITE-ERR-LINE.
  MOVE SPACES TO ERR-LINE.
  MOVE '' TO ERR-CC.
  PERFORM U200-WRITE-ERR-LINE.
IDMS-ABORT.
  EXIT.
IDMS-ABORT-EXIT.
  COPY IDMS IDMS-STATUS.
```